

# Bridging the Gap in Infrastructure Modeling & Visualization

Benjamin Boughton, Brian Luu, Evan Jones,  
Jacob Shalkhauser, Mernegar Dorgoly, & Travis Cole

## I. Introduction

Within the United States, there are more than 617,000 bridges each of which is paramount to ensuring the safe and reliable transit of goods, services, and individuals throughout the country. Despite their critical function, a significant proportion of the nation's bridges are in a state of disrepair, creating an issue of substantial public concern. However, this issue is not often recognized in part due to the lack of non-proprietary solutions for transparent exploration and visualization of the topic. Today's most recognized tool to explore bridge data, the "InfoBridge" web portal [1], lacks the clarity and flexibility expected of modern web applications.

## II. Background & Problem Definition

This gap in existing technology creates the need for an open-source and user-friendly interface for both experts and non-experts alike to better visualize, understand, and explore both existing national bridge data [2] and deterioration modeling approaches. The problem and how we intend to solve it is two-fold, composed of enabling a unified modeling and visualization framework for the United States bridge inventory systems that improve upon present solutions. This improvement will take the form of an interactive, user-friendly web application that offers simple access to bridge infrastructure data, visualizations, and models.

### A. Visualization Survey

At present, the single freely available solution is the "InfoBridge" web portal [1]. The portal itself has data stored in a format not tailored to visualization design principles. Column names make data both inaccessible and confusing to expert and non-expert users alike and current geographic mapping lacks any interactivity or additional meaning beyond general location making the visualization uninformative. Performance and projection charts are visually unappealing and uninformative given the layout, size, and design of their associated sections thus limiting data exploration potential.

### B. Modeling Survey

Three general categories broadly encompass the present body of academic work in bridge condition modeling:

deterministic models, probabilistic models, and deep neural networks models. Deterministic models incorporate a set of defined relationships among features like time, age, region, etc. derived from human feature engineering and selection that would be used to model a given bridge component's need for replacement or repair based upon thresholds determined by engineers [3]. These initial approaches proved far too simplistic and inefficient for the task, leading to over replacement and poor resource allocation of components and labor. Gradually, their complexity increased through the adoption of linear regression models [4], polynomial regression models [5], and ensemble methods such as Random Forests [6]. However, their utility is limited by their rigid structure, susceptibility to multicollinearity, and reliance upon non-scaleable and often biased human feature engineering. Efforts have been made to remedy these shortcomings through variable selection methods such as LASSO and Step-wise Selection [7] as well as dimensionality reduction methods such as tensor decomposition [8]. Although effective in practice, this class of models is unable to quantify uncertainty with respect to inputs or predictions. This quality, which is key to risk assessment, is the primary reason for the creation of the probabilistic class of models. Presently the most utilized among these models are Markov chains [9] and as an extension of it, Bayesian Decision Networks [10] both of which permit the modeling of a bridge concerning its present known condition, a major improvement upon the static probabilistic models. Furthermore, similar to early generation models there is still significant subjective feature engineering and distributional approximation taking place in the construction of the transition probability matrices of these decision networks and Markov processes [11]. AI models directly addresses this issue as well as the necessity of expert feature engineering which plagues both deterministic and probabilistic models. This is most commonly achieved through the utilization of deep artificial neural networks. These computational models utilize several processing layers to learn and create a hierarchy of abstraction to best represent the training set [12]. A wide variety of neural network

architectures have been proposed from dense models [13] to convolutional networks [14] that more effectively engineer features through up-sampling of the feature space. Lesser known, but still relevant alternative hybrid AI methods include clustering approaches through KNN [15] and K-means [16] [6] as a feature engineering or preprocessing step to more effectively apply subsequent probabilistic modeling approaches. Furthermore, AI methods are far less transparent especially to non-expert users compared to the prior two classes of models, and still do not provide a complete solution to the temporal aspect of the problem. Despite the substantial body of work that composes present modeling approaches for bridge deterioration and forecasting, there is still significant room for improvement. The most prevalent short-coming which can be attributed to all contemporary modeling approaches is the lack of data integration across some of the most crucial known causes of bridge deterioration such as ambient temperature [17], climate [18], traffic, and accident data [19], and utilizing only a fraction of the national data available. Collectively, each of these historical shortcomings has motivated and shaped the development of our proposed improved methodology of bridge deterioration modeling.

### III. Methodology

The defining differences in our project’s methodology and the existing body of work within the field come down to four primary areas of innovations we plan to implement that collectively vastly improve the state of the art. Firstly, our new approach to infrastructure design and integration allows for access by both desktop and mobile users with significant speed ups created by our back-end infrastructure. Secondly, our expanded data aggregation, consolidation, and cleaning efforts borrow from the best of the existing literature base while expanding to promising new features such as weather & traffic. Similarly, our modeling approach also benefits from this more expansive feature space, and when combined with more modern and refined model architectures, ultimately allows more comprehensive models and more accurate predictions. Finally, our new visualization methodology adds a new layer of potential for data exploration and presentation via our central focus on interactivity. A further discussion of our project’s innovations in methodology are discussed in the following sections.

#### A. Infrastructure Methodology

Robust and flexible infrastructure is key to the operation

and accessibility of our tool. With the project’s emphasis on easy data exploration and accessibility, our team recognized the importance of creating an open-source and fast tool. As such, we decided to use Python’s Flask web framework to host the application’s web interface, a MySQL database for data storage, and Tableau for our visualizations. We will use Amazon Web Services (AWS) to bring these pieces together, with an Elastic Compute Cloud (EC2) instance for application hosting, a Relational Database Service (RDS) MySQL instance to serve the database, and a Simple Storage Service (S3) bucket to store files. **Figure 1** details the infrastructure. Brought into the EC2 instance through an HTTP request, calls will be made to the MySQL database and the S3 bucket to retrieve data and files. Data will also be tunneled through Tableau Online, which will produce the visualizations embedded into the application’s HTML pages. The Flask framework offers simplicity, giving us detailed control over what we add to our application and how it’s presented. With Flask serving HTML, we have the capabilities of HTML, CSS, JavaScript, etc. to fully customize the user-interface. This leads to innovation that goes beyond the existing tools: with Flask we can make both desktop and mobile interface layouts, integrate JavaScript and D3, and provide a lightweight tool that works on all browsers. We decided on a MySQL database as it is fast, reliable, and open-source. Packages such as SQLAlchemy make it simple to write/read data directly via Python scripts, which is necessary for our Flask-based application. MySQL also easily connects to Tableau Online, the visualization tool we will use to present our data. At our infrastructure’s core, AWS grants us configurability and near-perfect uptime, allowing us to focus on features, visuals, and models. By hosting on EC2, we do not need to manage a physical server, making it simple to host our application on the web. By using an MySQL RDS instance, we don’t need to do configuration beyond adding data objects. The same goes for the S3 bucket, all the configuration is already complete. The major benefit of using these AWS services is that we can quickly scale depending on our needs. If we need more than 20GB of storage space in our RDS instance, we can quickly increase our storage without downtime. Likewise, we can increase the computing power of our Flask application as functionality requirements increase. As projected in our proposal, AWS costs still remain at \$0 due to the AWS credits generously allocated for the course.

## B. Data Preparation Methodology

The first step in the data workflow was the aggregation of several data sets we utilized for modeling and visualization purposes later in the report. Firstly, all data concerning the location, status, and condition of bridges from years 1993-2020 was pulled from the National Bridge Inventory database [2]. Monthly traffic volume and accident data was collected from 1993-2020 via the FHA traffic volumes database [20], and was linked to corresponding bridges via route and regional encoding information. Finally, weather data was collected from the NOAA monthly weather database [21] utilizing the report stations' latitudes and longitudes to match with bridges. Since clean and accurate data is the cornerstone to any thorough analysis, we spent a significant amount of time preparing and cleaning the dataset. First was the consolidation and merging of 27 years of independently stored bridge data. Inconsistent data conventions across years added complications, leading us to make manual adjustments. Likewise, there were instances of duplicated bridge records that had to be reconciled. Additionally, since most of the data was coded according to internal numeric NBI conventions, we converted this data to human-readable text. Due to our focus on the 48 contiguous U.S. states, bridges presiding in territories, Alaska, and Hawaii were removed. After cleaning the bridge data set, we began matching each bridge with a corresponding NOAA weather station based on the Haversin distance formula which is used to calculate distances between given latitude and longitude endpoints [22]. By pairing each bridge with the closest station, we save storage space and provide more accurate information about each bridge's unique exposure to weather and climate. Next, we began converting the FHA traffic volumes data set into matching fields for each bridge by year. For cleaning, we only had to identify and remove outliers. After cleaning operations were complete, the data was made available to the visualization and modeling teams via our MySQL database. Here, various virtual tables and indexes were constructed to permit fast querying for common requests and queries.

## C. Modeling Methodology

After completion of data aggregation and cleaning, we began the data pre-processing necessary for modeling. To optimize our ability to properly model future bridge conditions, it became clear that it makes the most sense in terms of accuracy to divide the U.S. into nine geographic regions instead of trying fit one model for

all U.S. bridges. Then, after filtering all the bridges for a given geographical region, we would then fit a model on just that subset, repeating this process on the remaining 8 regions. Noticing this improvement in model ability compared to a nationwide model, we reviewed additional literature and discovered that particle swarm and K-means clustering research methods [23] had explored this effect in great detail. This has much to do with the inherently large differences in weather, climate, and landforms that encompass the diverse geography of the U.S. By creating regional subsets, we were able to better isolate important factors specific to the bridges within each region. For each regional subset, we will conduct feature selection and dimensionality reduction experimenting with different selection criteria and methods. As the feature space for this problem is vast, this took a significant proportion of the modeling team's time. Many of the features required significant pre-processing due high cardinality among some predictors. Decisions were made on their inclusion or removal in order to reduce the effect of the curse of dimensionality. Additionally, several features contained a large proportion of null values which would be uninformative and confounding to modeling efforts, and were thus removed. When preprocessing is complete, we will advance into the modeling experimentation and evaluation phase. After all models have been fit, we will divide the full dataset into regional subsets again; according to their respective model, we will run inference on the full data set offline. These predictions will be sampled and fed through a test-bed constructed by the modeling team to verify logical consistency and validity of predictions. When the resulting predictions have been validated by the test-bed, they will then be stored within the database for later retrieval and visualization.

## D. Visualization Methodology

For our visualizations, we will be utilizing Tableau [24], hosted on Tableau Public to embed dashboards within our application. The visualization methodology will focus on presenting bridge information on a map of the continental U.S. identified by GPS coordinates using a choropleth map. Bar charts alongside this will display key information about the bridges and provide modeling results. These will be interactive visualizations, by use of filters and location selectors on the map. We will use Tableau's tool-tip feature to further display detailed bridge records, statistics, and other relevant information such as historical weather

trends and traffic patterns. This data will be read from our MySQL database. The bulk of the visualization development will take place during the timeline's later-half, as it relies on the data and model results. With Tableau, we will be able to quickly develop high-quality visualizations and iterate during the design process. We have created sample visualizations with our expected data volume to test the performance of the Tableau dashboard embedded into our application.

#### **IV. Upcoming Experiments & Evaluation**

The primary area of experimentation and evaluation concerns the construction and refining of our bridge condition models. We began work on constructing several basic models from each of the three primary model classes discussed in the literature review in order to prototype various approaches. Using basic models as a baseline, we can then make more informed decisions on which models show promise for further optimization and hyper parameter tuning via grid-search. After conducting this process for each of the three model classes and selecting two of the best from each class, the resulting group of models will then be evaluated against one another via n-fold cross-validation. This cross-validation will then yield our best "production" model for usage with that given geographical region. The process will then be repeated eight additional times thus creating nine different models for each of the nine geographic regions. The experimentation process mentioned above is still ongoing, and too early to present any concrete results or figures. For this reason the rest of this section and next will remain under construction as we conduct our remaining experiments and perform our evaluation of the results.

#### **V. Discussion of Results**

##### **SECTION UNDER CONSTRUCTION**

#### **VI. Plan of Activities & Work Distribution**

In our proposal, we noted that our team is broken into the data, modeling, and visualization sub-teams. Ben and Jacob of the data team being responsible for cleaning/integrating data and database maintenance, Evan and Mernegar of the modeling team being responsible for modeling and back-end optimization, and Brian and Travis of the visualization team being responsible for creating visualizations. We also noted we'd measure progress in deliverable checkpoints at 3 and 6 weeks. This sub-team breakdown remains at this time in the process. Ben is finalizing data integration into the MySQL database, keeping tables accurate and optimized; this work is estimated to take 20 more hours. Jacob is setting up the AWS infrastructure and building

the Flask application, which involves designing the web interfaces/routing and integrating visualizations; this is estimated to take an additional 30 hours. Evan and Mernegar are conducting additional data pre-processing and feature selection to prep modeling and analysis. There is an estimated 12 hours of work still left to complete. Additionally, Evan is working on training Markov chain and Neural network models while Mernegar trains regression, SVM, and random forest models. We estimate 30 hours of modeling work remaining on these tasks. On the visualization sub-team, Travis will verify and arrange the availability and access of data to Tableau services. Brian will spend 10 hours to parse the received data and make any adjustments for the visualizations. Brian and Travis will work together for 40 hours to identify critical information for use in tool tips. This team will spend another 40 hours to make final adjustments. All team members will work together on the last week of November for the final report, focusing on the Discussion of Results and Conclusion sections. Collectively, all team members have contributed a similar amount of effort.

#### **VII. Conclusion**

As our project development continues, we remain confident in our goal of using advanced modeling and interactive visualization techniques to improve upon the existing solutions in this space. We have assembled some of today's most innovation software solutions (e.g. Flask, Tableau, AWS, etc.) and most advanced modeling techniques to begin the creation of an application that will bring clear and accurate data on bridge infrastructure to all individuals in the United States. Our data sub-team has created a database of critical datasets and designed a reliable infrastructure for a user-friendly Flask application. The visualization team has begun to build interesting and interactive visualizations using Tableau's public tools, and remains confident in their ability to expertly present our data in valuable ways. On the modeling side, preliminary analysis has shown that the task as formulated lends itself well to modeling. We are optimistic about our informal modeling results and are excited for the value the final results will bring to our application. Overall, we have already begun to see the increasing inherent value in our application and we look forward to what it will bring in the weeks ahead.

## REFERENCES

- [1] F. H. A. . N. B. Inventory, "Ltpb infobridge," <https://infobridge.fhwa.dot.gov/Data>, U.S. Department of Transportation, 2021, accessed: 2021-10-11.
- [2] —, "National bridge inventory dataset," <https://infobridge.fhwa.dot.gov/Data/Dashboard>, U.S. Department of Transportation, 2021, accessed: 2021-10-11.
- [3] W. A. Hyman and D. J. Hughes, *Computer model for life-cycle cost analysis of statewide bridge repair and replacement needs*. National Research Council Transportation Research Board, 1983, no. 899.
- [4] R. Goyal, M. J. Whelan, and T. L. Cavalline, "Characterising the effect of external factors on deterioration rates of bridge components using multivariate proportional hazards regression," *Structure and Infrastructure Engineering*, vol. 13, no. 7, pp. 894–905, 2017. [Online]. Available: <https://doi.org/10.1080/15732479.2016.1217888>
- [5] M. Bolukbasi, J. Mohammadi, and D. Ardit, "Estimating the future condition of highway bridge components using national bridge inventory data," *Practice Periodical on Structural Design and Construction*, vol. 9, no. 1, pp. 16–25, 2004. [Online]. Available: <https://ascelibrary.org/doi/abs/10.1061/%28ASCE%291084-0680%282004%299%3A1%2816%29>
- [6] B. A. Bektas, A. Carriquiry, and O. Smadi, "Using classification trees for predicting national bridge inventory condition ratings," *Journal of Infrastructure Systems*, vol. 19, no. 4, pp. 425–433, 2013. [Online]. Available: <https://ascelibrary.org/doi/abs/10.1061/%28ASCE%29IS.1943-555X.0000143>
- [7] J. Wettach-Glosser, A. Unnikrishnan, and T. Schumacher, "Survival analysis of concrete highway bridge decks in oregon utilizing lasso and stepwise-variable selection," *Journal of Bridge Engineering*, vol. 25, p. 04020077, 10 2020.
- [8] O. Adarkwa, T. Schumacher, and N. Attoh-Okine, "Multiway analysis of bridge structural types in the national bridge inventory (nbi): A tensor decomposition approach," in *2014 IEEE International Conference on Big Data (Big Data)*. IEEE, 2014, pp. 1–6. [Online]. Available: [10.1109/BigData.2014.7004423](https://doi.org/10.1109/BigData.2014.7004423)
- [9] M. Cesare, J. Santamarina, C. Turkstra, and E. Vanmarcke, "Modeling bridge deterioration with markov chains," *Journal of Transportation Engineering-asce - J TRANSP ENG-ASCE*, vol. 118, 11 1992.
- [10] D. Straub, "Stochastic modeling of deterioration processes through dynamic bayesian networks," *Journal of Engineering Mechanics*, vol. 135, no. 10, pp. 1089–1099, 2009. [Online]. Available: <https://ascelibrary.org/doi/abs/10.1061/%28ASCE%29EM.1943-7889.0000024>
- [11] S. Madanat, R. Mishalani, and W. H. W. Ibrahim, "Estimation of infrastructure transition probabilities from condition rating data," *Journal of Infrastructure Systems*, vol. 1, no. 2, pp. 120–125, 1995. [Online]. Available: <https://ascelibrary.org/doi/abs/10.1061/%28ASCE%291076-0342%281995%291%3A2%28120%29>
- [12] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, pp. 436–44, 05 2015.
- [13] Y.-H. Huang, "Artificial neural network model of bridge deterioration," *Journal of Performance of Constructed Facilities*, vol. 24, no. 6, pp. 597–602, 2010. [Online]. Available: <https://ascelibrary.org/doi/abs/10.1061/%28ASCE%29CF.1943-5509.0000124>
- [14] H. Liu and Y. Zhang, "Bridge condition rating data modeling using deep learning algorithm," *Structure and Infrastructure Engineering*, vol. 16, no. 10, pp. 1447–1460, 2020. [Online]. Available: <https://doi.org/10.1080/15732479.2020.1712610>
- [15] R. Assaad and I. H. El-adaway, "Bridge infrastructure asset management system: Comparative computational machine learning approach for evaluating and predicting deck deterioration conditions," *Journal of Infrastructure Systems*, vol. 26, no. 3, p. 04020032, 2020. [Online]. Available: <https://ascelibrary.org/doi/abs/10.1061/%28ASCE%29IS.1943-555X.0000572>
- [16] M. Radovic, O. Ghonima, and T. Schumacher, "Data mining of bridge concrete deck parameters in the national bridge inventory by two-step cluster analysis," *ASCE-ASME Journal of Risk and Uncertainty in Engineering Systems, Part A: Civil Engineering*, vol. 3, no. 2, pp. 1–17, 2017. [Online]. Available: <https://ascelibrary.org/doi/abs/10.1061/AJRUA6.0000889>
- [17] S. Sharma and S. Sen, "Bridge damage detection in presence of varying temperature using two-step neural network approach," *Journal of Bridge Engineering*, vol. 26, no. 6, p. 04021027, 2021. [Online]. Available: <https://ascelibrary.org/doi/abs/10.1061/%28ASCE%29BE.1943-5592.0001708>
- [18] A. Nasr, E. Kjellström, I. Björnsson, D. Honfi, O. L. Ivanov, and J. Johansson, "Bridges in a changing climate: a study of the potential impacts of climate change on bridges and their possible adaptations," *Structure and Infrastructure Engineering*, vol. 16, no. 4, pp. 738–749, 2020. [Online]. Available: <https://doi.org/10.1080/15732479.2019.1670215>
- [19] R. Cao, A. K. Agrawal, S. El-Tawil, and W. Wong, "Performance-based framework for evaluating truck collision risk for bridge piers," *Journal of Bridge Engineering*, vol. 25, no. 10, p. 04020082, 2020. [Online]. Available: <https://ascelibrary.org/doi/abs/10.1061/%28ASCE%29BE.1943-5592.0001618>
- [20] F. H. Administration, "Federal highway administration traffic volumes dataset," <https://www.fhwa.dot.gov/policyinformation/tables/tmasdata/>, U.S. Department of Transportation, 2021, accessed: 2021-10-11.
- [21] NOAA, "Noaa: Daily weather records," <https://www.ncdc.noaa.gov/cdo-web/datatools/records>, NOAA, 2021, accessed: 2021-10-11.
- [22] Z. Arifin, M. R. Ibrahim, and H. R. Hatta, "Nearest tourism site searching using haversine method," in *2016 3rd International Conference on Information Technology, Computer, and Electrical Engineering (ICITACEE)*, 2016, pp. 293–296.
- [23] S. Galvan-Nunez and N. Attoh-Okine, "Hybrid particle swarm optimization and k-means analysis for bridge clustering based on national bridge inventory data," *ASCE-ASME Journal of Risk and Uncertainty in Engineering Systems, Part A: Civil Engineering*, vol. 3, no. 2, p. F4016001, 2017. [Online]. Available: <https://ascelibrary.org/doi/abs/10.1061/AJRUA6.0000864>
- [24] "What is tableau?" [Online]. Available: <https://www.tableau.com/why-tableau/what-is-tableau>

APPENDIX I

Fig. 1. Project Modeling Diagram

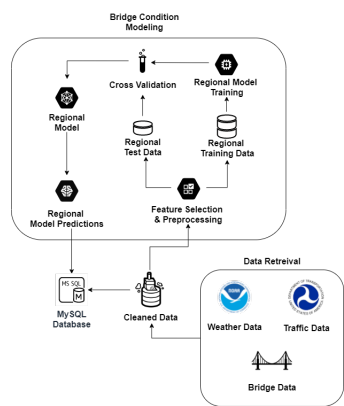


Fig. 2. Project Infrastructure Diagram

